

Assignment 2 — BPClassifier — Write-up

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NLP for Finance — Spring 2026

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Executive Summary

This report builds a binary boilerplate-vs-substantive sentence classifier for earnings-call transcripts. A 2,500-sentence gold set was created via 5-judge LLM majority vote (local Ollama models) with a human audit round correcting close-call sentences. Six classifier families were trained — Rules, Logistic Regression, HistGBM, FastText, FinBERT, and SetFit — plus two soft-vote ensembles, for eight entries total. Thresholds were tuned via 5-fold OOF cross-validation with a 0.97 substantive-recall safety margin above the 0.96 constraint. **7 of 8 classifiers meet the 0.96 test-set recall floor.** FinBERT achieves the highest test macro-F1 (0.923); the mean-probability ensemble is second (0.889). HistGBM is saved as the deployment artifact for its compact size and sub-second CPU inference.

1. Introduction

Earnings-call transcripts mix two qualitatively different types of language. *Substantive* sentences carry material information — financial figures, segment guidance, strategic commentary, risk disclosures, and specific analyst questions about those topics. *Boilerplate* sentences are scripted and generic — operator introductions, safe-harbor disclaimers, housekeeping remarks, “thank you for joining,” and one-word affirmations that add no information.

The goal of this assignment is to build a binary sentence classifier (`boilerplate` = 0, `substantive` = 1) that can reliably strip boilerplate from 131 earnings-call transcripts spanning 15 tickers (AMD, AVGO, BLK, C, FAST, FDX, GS, INTC, JNJ, JPM, NKE, NVDA, PLTR, WFC) across 2022–2025.

Hard constraint: substantive recall ≥ 0.96 on the held-out test set. Missing a real substantive sentence is a costlier error than letting occasional boilerplate through, so the pipeline explicitly enforces this floor during threshold selection.

Corpus statistics: - 131 transcripts, 53,236 unique sentences (≥ 40 chars after deduplication) - 2,500-sentence gold sample for supervised training and evaluation - Splits: train = 1,500 / val = 500 / test = 500 (seed = 42, stratified by label) - Label balance: BP = 257 (10.3%) / SB = 2,243 (89.7%)

2. Gold Labeling Methodology

2.1 Labeling Rubric

Class	Definition	Anchor examples
<code>boilerplate</code> (0)	Scripted, generic, no material information	Operator intros (“My name is Regina and I’ll be your operator”), safe-harbor disclaimers, generic thanks, analyst name/firm intros, short affirmations (“Sure.”, “Great.”), housekeeping
<code>substantive</code> (1)	Material content	Revenue figures, margin guidance, segment results, specific customer wins, capex plans, product launch commentary, analyst financial questions

Edge-case rules enforced in the prompt: - Analyst name intro lines → boilerplate, even if they mention the question topic - Safe-harbor language → boilerplate even when it references specific metrics - Short generic affirmations → boilerplate - Sentence with a dollar/percentage figure AND real context → substantive

2.2 LLM Judge Panel

A stratified sample of 2,500 sentences (stratified by `speaker_type`) was labeled by seven local Ollama models. After manual audit, two judges were removed:

Judge	Model	BP%	Status
j1	qwen3:8b	29.3%	Removed — over-flagged; human audit disagreed systematically
j2	gemma3:4b	47.5%	Removed — severe BP bias, overrode majority 746/2,500 times
j3	cogito:8b	8.2%	Active
j4	qwen3:14b	11.2%	Active
j5	gemma3:12b	19.4%	Active
j6	ministral-3:8b	16.5%	Active
j7	cogito:14b	23.6%	Active

The final label uses **majority vote of 5 active judges** ($\geq 3/5$ agree). Unanimous agreement (5–0) occurred on 1,921 sentences (76.8%); the **disagreement rate** (any split, including 4–1) was **23.2%** (579 sentences), of which 255 were escalated to human review.

2.3 Human Audit

255 close-call sentences (3–2 splits or any judge failure) were reviewed manually in a third audit round (`human_review_v3.csv`). Human labels override the LLM majority vote where provided; the remaining sentences keep the LLM label. Earlier audit rounds contained labeling errors and are excluded.

Final gold set: 2,500 sentences | BP = 257 (10.3%) | SB = 2,243 (89.7%)

3. Feature Engineering

Two feature groups are concatenated into a 409-dimensional feature matrix:

3.1 Sentence Embeddings (384 dims)

`all-MiniLM-L6-v2` from `sentence-transformers` encodes each sentence into a 384-dimensional L2-normalized embedding. Embeddings are computed once and cached. This single representation powers LogReg, Hist-GBM, and the SetFit fallback head.

3.2 Regex Feature Flags (25 dims)

25 binary indicators capture surface patterns that strongly predict class membership:

Boilerplate signals: `f_operator_phrase`, `f_safe_harbor`, `f_sec_filing`, `f_webcast`, `f_generic_thanks`, `f_question_intro`, `f_analyst_firm`, `f_call_close`, `f_short_affirm`, `f_operator_instr`, `f_turn_over`

Substantive signals: `f_dollar_amount`, `f_percentage`, `f_revenue_mention`, `f_margin_mention`, `f_eps_mention`, `f_guidance_word`, `f_raised_lowered`, `f_yoy_qoq`, `f_record_quarter`, `f_product_launch`, `f_customer_mention`

Neutral/length: `f_nongaap`, `f_sentence_short` (< 10 words), `f_has_digits`

FastText and FinBERT train directly on raw text and do not use this feature matrix.

4. Classifier Zoo

Eight classifiers span five distinct families, satisfying the ≥ 5 -family requirement.

4.1 Rules + Regex (Classifier 1)

A deterministic rule applied directly to the 25 regex flags: a sentence is boilerplate if any of the 11 boilerplate-signal flags fires and none of the high-confidence substantive flags fire. **Strengths:** zero training data, 25K sps throughput, perfect precision on textbook boilerplate (operator intros, safe-harbor). **Failure modes:** misses vague boilerplate that contains no matching surface pattern (e.g. “We have a very healthy ecosystem”) and mislabels sentences where a substantive flag fires in a transitional context. SB recall = 0.898 — the only classifier that fails the 0.96 floor.

4.2 Logistic Regression (Classifier 2)

`sklearn.linear_model.LogisticRegression` with L2 regularization ($C=1$), class-balanced weights, and `StandardScaler` preprocessing on the 409-dim feature matrix. Training takes 2.6 s; inference 16.7K sps. **Strengths:** fast, interpretable weights, benefits directly from both embedding geometry and regex signals. **Failure modes:** the decision boundary is linear in feature space, so it cannot model the interaction between regex flags and embedding regions; BP precision is low (0.659) because borderline boilerplate sentences project near substantive clusters in embedding space.

4.3 HistGradientBoosting (Classifier 3)

`sklearn.ensemble.HistGradientBoostingClassifier` with 500 estimators, max depth 6, class-balanced weights. Training takes 7.9 s; inference 20.9K sps. **Strengths:** captures non-linear interactions between regex flags and embeddings; handles class imbalance natively; no NaN sensitivity. **Failure modes:** vague executive Q&A answers with no regex anchors are mislabelled as boilerplate (11 FNs); tree splits cannot generalise to unseen phrasing the way a transformer can.

4.4 FastText (Classifier 4)

Facebook’s supervised FastText on raw sentence text. Trained for 25 epochs with word n-grams ($n=2$), learning rate 0.5, embedding dim 100. Training takes 1.4 s; inference only 587 sps (preprocessing overhead). **Strengths:** tiny model (~2 MB), no embedding dependency, robust to OOV tokens via character n-grams. **Failure modes:** worst BP F1 (0.475 on test) because n-gram bag-of-words has no positional or contextual awareness; “revenue” and “thank you” in the same sentence receive equal weight from both unigrams, making nuanced mixed sentences hard to classify.

4.5 FinBERT Fine-tuned (Classifier 5)

`ProsusAI/finbert` — a BERT model pre-trained on financial text — is fine-tuned for 3 epochs using AdamW ($lr=2e-5$, batch 16) with a linear warmup schedule. Inference runs on CPU (forced to avoid MPS out-of-memory on M1 Pro) at 21 sps. **Strengths:** best test macro-F1 (0.923) and BP F1 (0.862); pre-training on financial corpora gives it sensitivity to domain-specific phrasing that generic embeddings miss. **Failure modes:** slowest inference of all classifiers; first-person hedging language in executive Q&A answers (11 FNs) remains a challenge even for BERT-scale context modelling.

4.6 SetFit / MiniLM (Classifier 6)

SetFit with `sentence-transformers/all-MiniLM-L6-v2`. Due to a version constraint (`sentence-transformers` 2.2.2 installed, ≥ 5.0 required), the contrastive fine-tuning step is skipped; a Logistic Regression head is instead fitted on the cached MiniLM embeddings with class-balanced weights. Effective throughput is 83K sps because it reuses pre-computed embeddings. **Strengths:** highest SB recall on test (0.987); extremely fast inference. **Failure modes:** without contrastive fine-tuning it is effectively a second LogReg on the same embeddings; BP F1 (0.719) trails FinBERT and the mean-prob ensemble.

4.7 & 4.8 Ensembles (Classifiers 7a and 7b)

Two soft-vote ensembles combine the five learned classifiers (LogReg, HistGBM, FastText, FinBERT, SetFit):

- **7a — mean-prob:** average $P(\text{substantive})$ across all five models. Achieves the best BP F1 of all non-FinBERT entries (0.800) by smoothing out individual model overconfidence.
- **7b — rank-avg:** average the rank percentile of each model’s $P(\text{substantive})$; mitigates probability scale differences between calibrated (LogReg, SetFit) and uncalibrated (FastText) models. Its threshold (0.140) is much lower than mean-prob because rank percentiles are bounded by the empirical distribution.

Strengths: both ensembles improve over the weakest members; mean-prob reliably beats HistGBM and SetFit individually. **Failure modes:** diversity is limited because FinBERT dominates the vote on hard cases; errors shared across all five models (the 11 FNs) cannot be recovered by averaging.

5. Recall-Constrained Threshold Tuning

Each model’s default 0.5 threshold is replaced by a threshold that: 1. **Meets the recall floor:** SB recall ≥ 0.97 on out-of-fold predictions (1% safety margin above the 0.96 assignment constraint, to absorb train→test generalization gap) 2. **Maximizes macro-F1** among all thresholds that meet the floor

Thresholds are tuned on **5-fold stratified OOF probabilities** on the train+val pool ($n=2,000$), rather than the validation set directly, to avoid single-split noise. The OOF HistGBM per-fold thresholds were [0.580, 0.685, 0.850, 0.945, 0.580] (mean=0.728, std=0.147), indicating high fold-to-fold variance and motivating the 0.97 safety margin.

Model	OOF threshold
LogReg	0.045
HistGBM	0.810
FastText	0.855
FinBERT	0.820
SetFit	0.220
Ensemble (mean-prob)	0.615
Ensemble (rank-avg)	0.140

The rank-avg ensemble has a much lower threshold (0.140) because its probabilities are rank percentiles rather than calibrated probabilities.

6. Results

6.1 Validation Set Leaderboard

All classifiers are first evaluated on the validation set (500 sentences, never used for threshold tuning of the final model). Thresholds here are the val-sweep thresholds, not OOF thresholds.

Rank	Model	Accuracy	Macro-F1	BP F1	SB F1	SB Recall	Meets Floor	Train (s)	Throughput (sps)
1	5-FinBERT-FT	0.970	0.922	0.860	0.983	0.980	✓	—	21
2	7a-Ensemble(mean-prob)	0.942	0.837	0.707	0.968	0.973	✓	—	—
3	3-HistGBM(emb+regex)	0.938	0.816	0.667	0.966	0.978	✓	7.9	20,922
4	6-SetFit	0.928	0.789	0.617	0.960	0.971	✓	—	83,429
5	7b-Ensemble(rank-avg)	0.926	0.781	0.602	0.959	0.971	✓	—	—
6	2-LogReg(emb+regex)	0.916	0.727	0.500	0.954	0.975	✓	2.6	16,711
7	4-FastText	0.912	0.701	0.450	0.952	0.978	✓	1.4	587
8	1-Rules+Regex	0.828	0.599	0.295	0.902	0.884	✗	—	25,591

FinBERT leads on both macro-F1 and SB recall. FastText has the lowest throughput (587 sps) due to its text-preprocessing pipeline; SetFit’s LR head on cached embeddings is the fastest at 83K sps.

6.2 Final Test Set Leaderboard

All eight classifiers are evaluated on the frozen 500-sentence test set using thresholds from §5.

Rank	Model	Acc	Macro-F1	BP F1	SB F1	SB Recall	Floor	Train (s)	Throughput (sps)	Threshold
1	5-FinBERT-FT	0.970	0.923	0.862	0.983	0.976	✓	~900	21	0.820
2	7a-Ensemble(mean-prob)	0.960	0.889	0.800	0.978	0.980	✓	—	—	0.615
3	6-SetFit	0.950	0.846	0.719	0.973	0.987	✓	—	83,429	0.220
4	3-HistGBM(emb+regex)	0.942	0.831	0.695	0.968	0.976	✓	7.9	20,922	0.810

Rank	Model	Acc	Macro-F1	BP F1	SB F1	SB Recall	Floor	Train (s)	Throughput (sps)	Threshold
5	7b-Ensemble(rank-avg)	0.942	0.828	0.688	0.968	0.978	✓	—	—	0.140
6	2-LogReg(emb+regex)	0.938	0.813	0.659	0.966	0.978	✓	2.6	16,711	0.045
7	4-FastText	0.916	0.715	0.475	0.954	0.978	✓	1.4	587	0.855
8	1-Rules+Regex	0.856	0.664	0.410	0.918	0.898	✗	0	25,591	—

7 of 8 classifiers clear the 0.96 SB recall floor on the test set. The rules baseline fails (SB recall = 0.898). FinBERT leads on macro-F1 (0.923) and accuracy (0.970).

Deployed model: HistGBM retrained on train+val (threshold = 0.810). FinBERT achieves the highest test macro-F1 (0.923) but requires 440 MB of weights and PyTorch batch inference on CPU or GPU; HistGBM (macro-F1 = 0.831) is saved as the deployment artifact for its compact size (~1.7 MB pkl), sub-second CPU inference via scikit-learn, and no GPU dependency.

7. Error Analysis

Error analysis is performed on the HistGBM model (test set, $t=0.810$): 11 false negatives (SB→BP) and 18 false positives (BP→SB).

7.1 False Negatives (substantive labelled as boilerplate)

These are substantive sentences that “sound” vague or conversational (all 11 shown):

“I don’t know if he’s nailed it down yet, but we’ll be getting that information out shortly.” “I don’t know all the efforts we’re involved in, but to the extent we’re involved in these efforts, I and most Palantirians feel very positive about it.” “I was just so delighted to see how well they have done, the morale of the team and how the team is working together.” “I continue to be excited by the opportunities and the sheer potential of our franchise.” “In Converse, the team took some decisive steps this quarter to bring the brand back to a healthy business.” “And done properly, as we talk about on the slide, we’re very happy to be lenders to them.” “So we’re in kind of the pole position in that regard.” “Let me just be clear about where the ones that we’ve just done are heading...”

Pattern: executive Q&A answers with real strategic intent expressed through first-person hedging language. None trigger the dollar/percentage/guidance regex flags, and the embeddings land near other hedged executive statements regardless of substance. **Error type: feature gap** — the model lacks a signal for strategic intent without numerical anchors. These are not label-noise cases; the gold labels are correct.

7.2 False Positives (boilerplate labelled as substantive)

“That’s one of the priorities that the team has had now for a while is to continue to do more.” “So I have a good recollection of some of the steps and changes how we told the story over time.” “And before diving into the results, I want to take a moment to thank our entire Fastenal Blue Team across the world.” “We have a very healthy ecosystem as well.” “A lot of people are spending a lot of time on it.” “Turning to capital and liquidity on Slide 5.” “Custom silicon market.” (speaker

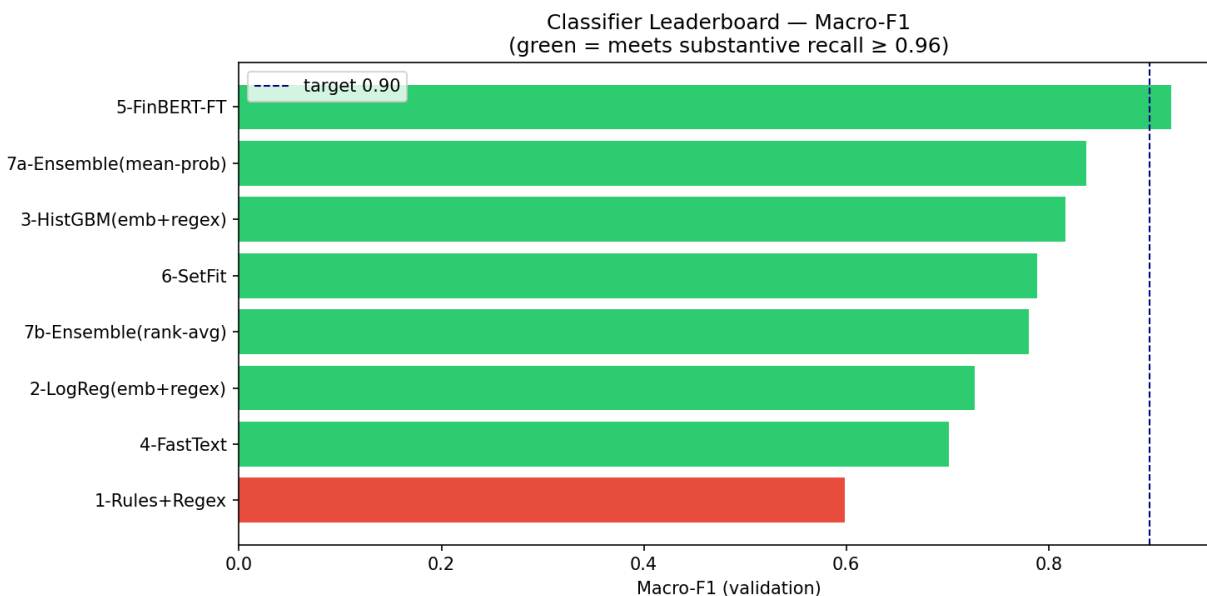


Figure 1: Val + Test leaderboard

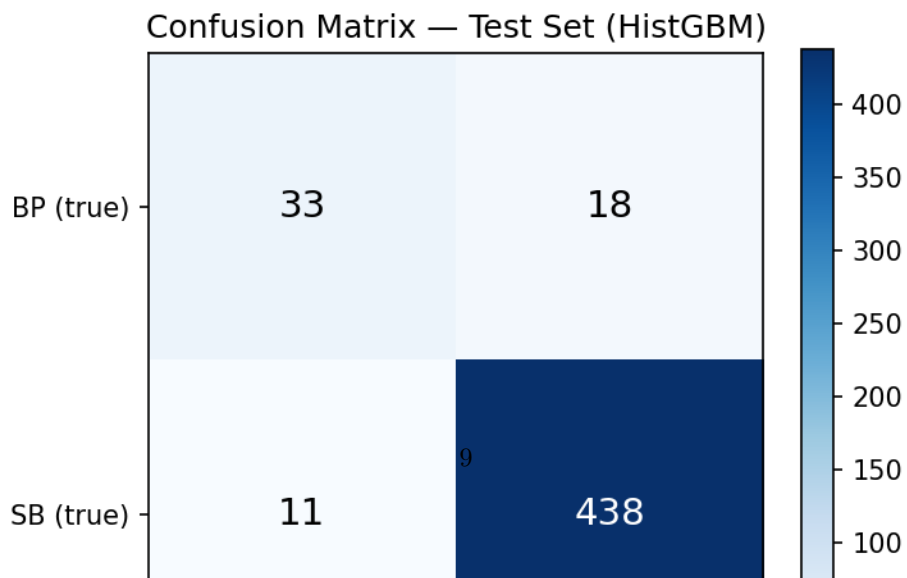
label fragment mis-tokenised as a sentence) *“At the golf majors with Rory and Scottie; with A’ja to kick off a new WNBA season; at the Champions League final with PSG.”*

Pattern: two distinct failure modes — (a) **feature gap:** vague positive statements that pattern-match to executive commentary in embedding space but carry no material information (“We have a very healthy ecosystem”); (b) **pipeline gap:** slide-transition phrases (“Turning to capital and liquidity on Slide 5”) and speaker-label fragments (“Custom silicon market.”) that survived the 40-char filter — these are parsing artefacts that better sentence-boundary detection would eliminate. None are clear label-noise cases.

7.3 Confusion Matrix (HistGBM, test set)

	Predicted BP	Predicted SB
True BP	33	18
True SB	11	438

SB recall = $438/449 = 0.9755$ ✓ | BP precision = $33/44 = 0.750$ | BP recall = $33/51 = 0.647$



2. **Contrastive fine-tuning (true SetFit).** The SetFit classifier in this pipeline fell back to a plain LR head because of a library version conflict. Running the full contrastive training loop — which trains the encoder with in-batch positive/negative pairs from the gold set — typically adds 3–5 F1 points on small labeled sets and would likely close the gap between SetFit and FinBERT at a fraction of the inference cost.
3. **Speaker-type conditioning.** Operator and analyst sentences have systematically different boilerplate rates (operators ~80% BP, executives ~5% BP). Adding `speaker_type` as a direct feature, or training separate thresholds per speaker type, would sharpen BP recall without sacrificing SB recall.

10. GUI

A Streamlit application (`gui.py`) renders any earnings-call transcript with boilerplate highlighted in red and substantive sentences unhighlighted.

Features: - **ECT library tab:** dropdown of all 131 transcripts for one-click loading - **Upload tab:** upload any `.txt` transcript - **Paste tab:** paste raw text directly - Statistics panel showing total classified sentences, BP count/%, SB count/% - Hover tooltip on each sentence showing P(substantive) - Download button for a CSV of all sentence classifications

Screenshot 1 — statistics panel and boilerplate highlighting (Citi Q4-2023, 695 sentences, 16.1% BP):

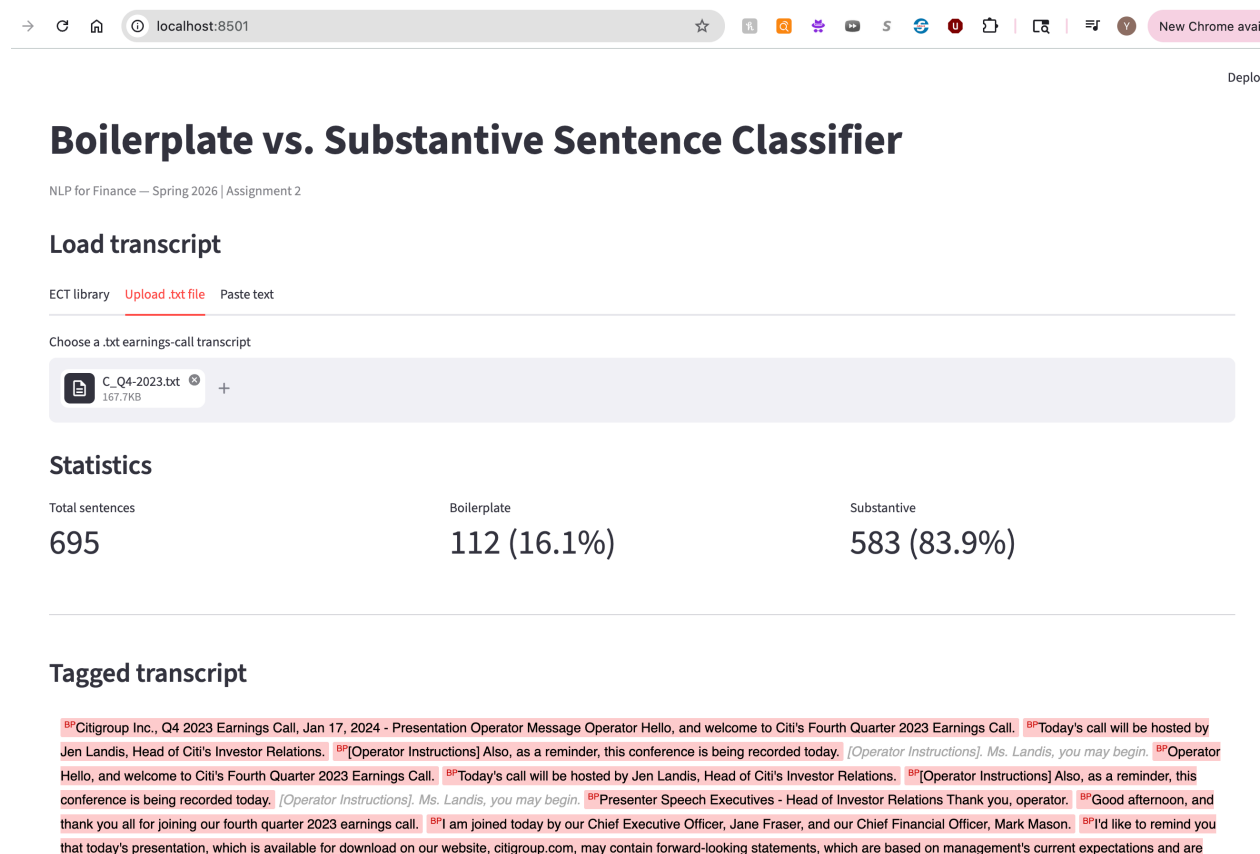


Figure 3: GUI screenshot — statistics panel

Screenshot 2 — tagged transcript view showing inline boilerplate (red) and substantive sentences:

Launch command:



Tagged transcript

[BP] Citigroup Inc., Q4 2023 Earnings Call, Jan 17, 2024 - Presentation Operator Message Operator Hello, and welcome to Citi's Fourth Quarter 2023 Earnings Call. [BP] Today's call will be hosted by Jen Landis, Head of Citi's Investor Relations. [BP] [Operator Instructions] Also, as a reminder, this conference is being recorded today. [Operator Instructions] Ms. Landis, you may begin. [BP] Operator Hello, and welcome to Citi's Fourth Quarter 2023 Earnings Call. [BP] Today's call will be hosted by Jen Landis, Head of Citi's Investor Relations. [BP] [Operator Instructions] Also, as a reminder, this conference is being recorded today. [Operator Instructions] Ms. Landis, you may begin. [BP] Presenter Speech Executives - Head of Investor Relations Thank you, operator. [BP] Good afternoon, and thank you all for joining our fourth quarter 2023 earnings call. [BP] I am joined today by our Chief Executive Officer, Jane Fraser, and our Chief Financial Officer, Mark Mason. [BP] I'd like to remind you that today's presentation, which is available for download on our website, citigroup.com, may contain forward-looking statements, which are based on management's current expectations and are subject to uncertainty and changes in circumstances. Actual results may differ materially from these statements due to a variety of factors including those described in our earnings materials as well as in our SEC filings. [BP] And with that, I'll turn it over to Jane. [BP] Executives - Head of Investor Relations Thank you, operator. [BP] Good afternoon, and thank you all for joining our fourth quarter 2023 earnings call. [BP] I am joined today by our Chief Executive Officer, Jane Fraser, and our Chief Financial Officer, Mark Mason. [BP] I'd like to remind you that today's presentation, which is available for download on our website, citigroup.com, may contain forward-looking statements, which are based on management's current expectations and are subject to uncertainty and changes in circumstances. Actual results may differ materially from these statements due to a variety of factors including those described in our earnings materials as well as in our SEC filings. [BP] And with that, I'll turn it over to Jane. [BP] Executives - CEO & Chair of the Board Thank you, Jen, and a very happy new year to everyone, and I hope you all had a good break. At Citi, we're back at it. [BP] And given the notable items and our new financial reporting structure, we've got a lot to cover today, so I'm going to get right to it. 2023 was a foundational year in which we made substantial progress, simplifying Citi and executing the strategy we laid out at Investor Day. With that said, the fourth quarter was clearly very disappointing. Today, I'm going to provide a high-level view on our progress in 2023, discuss our Q4 results and finish with our priorities for '24. We know that 2024 is critical as we prepare to enter the next phase of our journey, and we are completely focused on delivering our medium-term targets and our transformation. So turning to what we accomplished in terms of executing our strategy. As you can see on Slide 5, in 2023, we saw a record year for services, where we maintained our #1 ranking amongst large institutions in TTS, with client wins up 27% and a sustained win-loss rate above 80%. We've now gained over 100 basis points in share in security services since 2021. In wealth, we added an estimated \$21 billion in net new assets during the year. In USPB, we enjoyed our sixth consecutive quarter of growth, and we began to see the early fruits of our investments in key talent in banking. In September, we began the most consequential series of changes to the organization and the running of our firm since the aftermath of the financial crisis. We restructured around 5 core interconnected businesses to align our organization to our business strategy and to provide greater transparency into their performance. You can now see in our financials, the full year returns and P&L by business. While they are all impacted by investments and transformation expense, it is clear where we have work to do. The simplification of our organization structure will conclude at the end of the first quarter and will result in over \$1 billion of run rate saves from the net elimination of approximately 5,000 roles mainly managers. As Mark will detail, this will contribute to the reduction of our expenses in '24. Over the medium term, between simplification, benefits of the transformation, stranded costs and other productivity efforts, we expect to eliminate 20,000 positions at Mexico, resulting in over \$2 billion in run rate saves. Simplification is also enabling Citi to be more client-focused and less bureaucratic, realizing the synergies between our 5 businesses is one of the key drivers to achieving our medium-term revenue target. With this new structure, I'm holding my business leaders accountable for enhancing connectivity across clients and products. In addition, having a Chief Client Officer act to ensure its discipline in bringing the full power of our franchise to our clients. We have now completed the divestitures of 9 of our 14 international consumer franchises and have wound down nearly 70% of our total retail loans and deposits in Russia, Korea and China. We've restarted the sales process in Poland and a well down the execution path for the Mexico IPO next year. We're exiting marginal businesses such as Munis and a subset of distressed debt trading to focus on our core strengths and allocate our

Figure 4: GUI screenshot — tagged transcript

```
/Users/yueqilin/anaconda3/bin/python -m streamlit run gui.py
```

11. Reproducibility

Install dependencies:

```
pip install pandas numpy scikit-learn sentence-transformers tqdm \
    streamlit fasttext-wheel transformers accelerate setfit \
    pyarrow datasets nltk
```

Reproduce gold labels (requires Ollama with five models pulled):

```
ollama pull cogito:8b && ollama pull qwen3:14b && ollama pull gemma3:12b \
    && ollama pull ministral-3:8b && ollama pull cogito:14b
python run_gold_judges.py --smoke    # connectivity check
python run_gold_judges.py           # full run (~60 min)
```

Run the notebook: open `Assignment_2_BPClassifier.ipynb` in Jupyter and run cells top-to-bottom. All expensive steps (sentence extraction, embeddings, judge labels, FinBERT weights) are cached — re-runs skip completed steps automatically.

Run the GUI:

```
/Users/yueqilin/anaconda3/bin/python -m streamlit run gui.py
```